SKYE8 COMPANY LTD

Data Science and ML / AI Internship Track

FINAL PROJECT

Loan Collections Prioritisation

Issued to	NAHBILA FAITH
Roadmap reference	SKY8-DS-RM-005 (Core ML)
Project reference	SKY8-DS-FP-005
Duration	Seven days - issued Friday 25 September 2026, closes 23:59 Friday 2 October 2026
Repository	skye8-ds-fp-nahbila-faith
Submission	Skye8 Platform
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1. The Product

A microfinance institution has four hundred borrowers behind on payments and three collections officers. They work down the list by size of debt, which is the wrong order, because the largest debt is often the least recoverable and the small one that was going to be paid anyway wastes a visit.

You are building the prioritiser. It ranks the arrears book by who is most likely to pay if contacted this week, and recommends the contact method for each.

Why this needs machine learning. Recovery probability depends on how the borrower has behaved before, how long they have been late, what they borrowed for, and how they responded to previous contact. Those interact, and no officer can weigh them across four hundred accounts. A rule such as “biggest debt first” is exactly the mistake this replaces.

Who buys it. Every microfinance institution and credit union in Bamenda. Priced as a monthly licence, justified by recovery gained against officer time spent.

2. Data

Source	What it gives you
Home Credit Default Risk (Kaggle)	Free. Contains instalment payment history and prior credit behaviour, which is what you need to model recovery rather than origination.
LendingClub loan data	Free and widely mirrored. Full repayment histories including late payments and recoveries, with the eventual outcome of each loan.
Note on framing	You are not predicting default. You are predicting recovery given that default has already happened. That is a different target and a different population, and getting that distinction right is most of this project.

Day two checkpoint. Data loaded and the arrears population isolated, with its size and recovery rate reported.

3. The Work

Stage A - Define the target properly

- Write down exactly what counts as recovery and over what window. Thirty days, ninety days, partial payment - your choice, but state it and defend it.

- Isolate the population of accounts already in arrears. Training on all borrowers answers a different question.
- Report the base recovery rate. That is what happens if you do nothing.

Stage B - Features from behaviour

- Build features from payment history: how they paid before falling behind, how many days late, whether they have recovered from arrears before, the trend in their payments.
- Everything must be knowable at the moment the officer picks up the phone. A feature recorded after contact is not available to you.
- Write a test that fails if a post-contact field reaches the model.

Stage C - Model and rank

- Compare gradient boosting against a simple ordering by debt size. That ordering is the incumbent and you must beat it.
- Report precision at the top 50 and top 100 accounts, because that is how many the officers can actually work in a week.
- Report the money recovered under your ranking against the money recovered under biggest-debt-first. That figure is your business case.

Stage D - Say what to do, not just who

- Recommend a contact method per account - call, message, field visit - based on what has worked for similar borrowers.
- A field visit costs far more than a message. Weight the recommendation by cost, not only by recovery probability.

Stage E - Ship it

- Deploy a page where an officer sees this week's list, ranked, with a reason and a recommended action for each account.
- Include the expected recovery for the week so a manager can see what the list is worth.
- Be explicit about the human cost. These are real people in difficulty, and a collections tool that pushes officers to harass the vulnerable is a bad product. State what you would build in to prevent that.

5. What You Hand In

Five things. All five are required; a missing one is an incomplete submission.

Deliverable	Requirement
1. Working demo	A live URL anybody can open and use, or an installable APK. It must work on a phone. If it needs internet, say so; if it works offline, prove it.
2. Repository	Public GitHub repo under the Skye8 organisation. README that lets a stranger run it. No data files or credentials in git history.
3. Technical report	Four pages maximum. The problem, the data, what you built, how well it works, and what it gets wrong. Numbers, not adjectives.
4. Pitch	Five slides, five minutes, presented live. What it does, who buys it, why it is better than what they do now.
5. Business case	One page. Who pays, how much, what it costs to run per month, and how many customers before it breaks even. Use real figures you can source.

The business case is not decoration. If you cannot name a buyer in Bamenda and a price they would pay, you have built a demonstration, not a product.

6. Acceptance

1. The demo returns a ranked weekly list with reasons and recommended actions.
2. The target is recovery given arrears, not default, and the population is correctly isolated.
3. No post-contact information reaches the model, enforced by a test.
4. Performance is reported as precision at 50 and 100, not as overall accuracy.
5. Money recovered is compared against biggest-debt-first ordering.
6. Contact method recommendations account for cost.

The product is not the model. It is the difference between what the three officers recover with your list and what they recover without it, in francs.

7. Submission

Everything is submitted on the Skye8 platform. Nothing sent by email, WhatsApp or Drive link will be marked.

1. Submit the live demo URL or APK, the repository URL, and the commit hash of your final merge.
2. Attach the technical report, the slide deck and the business case.
3. Submit before 23:59 on Friday 2 October 2026. The platform timestamp is the only record.
4. Pitches are presented the following Monday evening in the online session.
